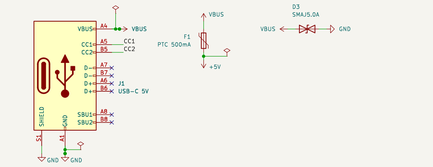
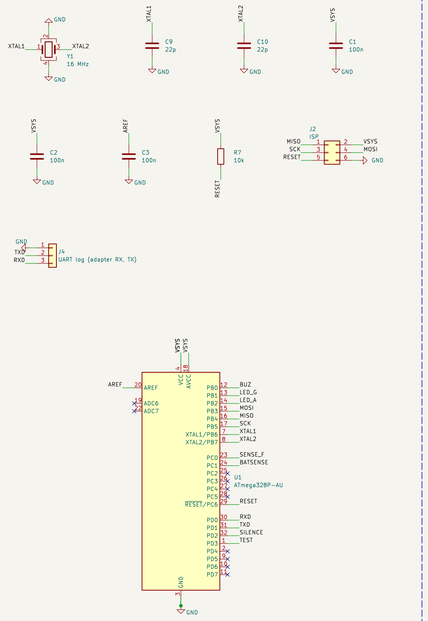
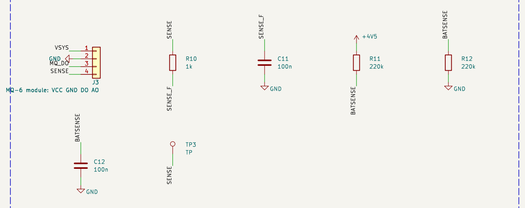
POWER: USB-C 5 V and 3xAA backup OR-ed into VSYS through Schottky diodes



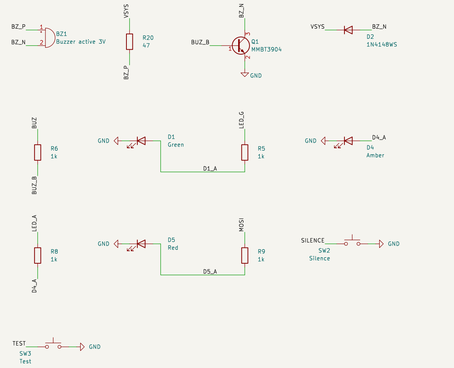
MCU: ATmega328P-AU, 16 MHz, ISP and UART header



SENSOR: MQ-6 module socket, filtered analog input, battery divider



ALARM: active buzzer, three LEDs, silence and test buttons



Gas-Suraksha PROTOTYPE / DATA-LOGGING BOARD, NOT A CERTIFIED GAS DETECTOR. It carries the alarm engine from firmware/lib/gas_core on an ATmega328P with a socket for a common MQ-6 module (VCC, GND, DO, AO; check your module's pin order). USB-C 5 V and a 3xAA pack are OR-ed through Schottky diodes into VSYS (about 4.1 to 4.7 V); the MQ-6 heater sees this, so calibrate readings per supply. The ACTIVE 3 V buzzer is fed through 47 ohm; the firmware switches pin D8 on and off. D11 (red LED) shares the MOSI line like an Arduino D13 LED. Program through the ISP header (firmware/release/gas_suraksha_demo.hex, 16 MHz, external crystal fuses); read logs on the UART header at 115200 baud. Do not sell or rely on it as a safety device.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /

File: gas_suraksha.kicad_sch

Title: Gas-Suraksha prototype and data-logging board (NOT a certified gas

Size: A1	Date: 2026-11-01	**Rev: A**
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